

Date	30 June 2026	Maximum Marks	4 Marks
Team ID	xxxx	Project Name	India's agricultural crop production analysis

PROJECT DESIGN PHASE-II

Solution Requirements (Functional & Non-functional)

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Ingestion & Multi-Source Integration	Integration with UPaG Portal API for live market pricing & advance production estimates.
		Ingestion of plot-level geotagged data from Digital Crop Surveys.
		Automated data sync with the Google Agricultural Understanding API for satellite-based field segmentation.
FR-2	Descriptive Analytics & Visualization	Interactive dashboards displaying state-wise and crop-wise (Rice, Wheat, Pulses) production output.
		Heatmaps for visual assessment of regional agricultural yields and cropping intensity.
		Time-series trend lines overlaying rainfall variations against historic crop productivity drop-offs.
FR-3	Predictive Yield Modeling	Machine learning engines to calculate expected yields based on historical FASAL data inputs.
		Stress and drought vulnerability forecasting utilizing geospatial indices.
FR-4	Reporting & Decision Support	Automated drafting and generation of weekly Crop Weather Watch Group (CWWG) summary insights.
		Custom filters allowing end-users to export granular datasets into standard PDF or Excel formats.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The interface must provide an interactive, scannable map-centric UI accessible to agritech developers, researchers, and government policymakers.
NFR-2	Security	Implementation of rigid Role-Based Access Control (RBAC) to isolate sensitive farmer plot boundaries and secure upstream API credential keys.
NFR-3	Reliability	Data accuracy must be cross-verified through a multi-source triangulation ledger mapping local crop-cutting tests to regional macro estimates.
NFR-4	Performance	Geospatial analytical queries and high-resolution map overlays must consistently render within 3 seconds under normal concurrent loads.
NFR-5	Availability	The tracking platform must guarantee a minimum uptime of 99.9% to maintain operational stability during critical sowing and harvesting spikes.
NFR-6	Scalability	The cloud-backed storage layer must automatically scale up to comfortably house expanding terabytes of high-resolution countrywide telemetry.